# Modular Arithmetic, Divisibility, Primes & Euler Totient

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Modular Arithmetic, Divisibility, Primes & Euler Totient Practice Exercises

**1** Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $35!$ in base $10$?

**a.** 7

**b.** 9

**c.** 8 ✓

**d.** 6

Since $10 = 2 \cdot 5$ and $v_2(35!) > v_5(35!)$, the number of trailing zeros equals

$v_5(35!) = \lfloor 35/5 \rfloor + \lfloor 35/25 \rfloor + \lfloor 35/125 \rfloor = 7 + 1 + 0 = 8$.

**2** Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $50!$ in base $10$?

**a.** 10

**b.** 12 ✓

**c.** 13

**d.** 8

Since $10 = 2 \cdot 5$ and $v_2(50!) > v_5(50!)$, the number of trailing zeros equals

$v_5(50!) = \lfloor 50/5 \rfloor + \lfloor 50/25 \rfloor + \lfloor 50/125 \rfloor = 10 + 2 + 0 = 12$.

**3** Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $65!$ in base $10$?

**a.** 15 ✓

**b.** 13

**c.** 16

**d.** 11

Since $10 = 2 \cdot 5$ and $v_2(65!) > v_5(65!)$, the number of trailing zeros equals

$v_5(65!) = \lfloor 65/5 \rfloor + \lfloor 65/25 \rfloor + \lfloor 65/125 \rfloor = 13 + 2 + 0 = 15$.

**4** Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $80!$ in base $10$?

**a.** 16

**b.** 20

**c.** 17

**d.** 19 ✓

Since $10 = 2 \cdot 5$ and $v_2(80!) > v_5(80!)$, the number of trailing zeros equals

$v_5(80!) = \lfloor 80/5 \rfloor + \lfloor 80/25 \rfloor + \lfloor 80/125 \rfloor = 16 + 3 + 0 = 19$.

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**5** Let $x$ be the smallest positive integer satisfying the simultaneous congruences $x \equiv 3 \pmod 5$ (whose initial progression is marked on the number line) and $x \equiv 6 \pmod 7$. Find $x$.

**a.** 18

**b.** 20

**c.** 13 ✓

**d.** 22

Write $x = 5k + 3$ and substitute into the second congruence: $5k + 3 \equiv 6 \pmod 7$. Solving modulo 7 gives the unique residue modulo $5 \cdot 7 = 35$, yielding smallest positive solution $x = 13$ (check: $13 = 2 \cdot 5 + 3$ and $13 = 1 \cdot 7 + 6$).

**6** Let $x$ be the smallest positive integer satisfying the simultaneous congruences $x \equiv 1 \pmod 5$ (whose initial progression is marked on the number line) and $x \equiv 7 \pmod 9$. Find $x$.

**a.** 21

**b.** 16 ✓

**c.** 25

**d.** 29

Write $x = 5k + 1$ and substitute into the second congruence: $5k + 1 \equiv 7 \pmod 9$. Solving modulo 9 gives the unique residue modulo $5 \cdot 9 = 45$, yielding smallest positive solution $x = 16$ (check: $16 = 3 \cdot 5 + 1$ and $16 = 1 \cdot 9 + 7$).

**7** Let $x$ be the smallest positive integer satisfying the simultaneous congruences $x \equiv 5 \pmod 7$ (whose initial progression is marked on the number line) and $x \equiv 1 \pmod 9$. Find $x$.

**a.** 19 ✓

**b.** 26

**c.** 28

**d.** 44

Write $x = 7k + 5$ and substitute into the second congruence: $7k + 5 \equiv 1 \pmod 9$. Solving modulo 9 gives the unique residue modulo $7 \cdot 9 = 63$, yielding smallest positive solution $x = 19$ (check: $19 = 2 \cdot 7 + 5$ and $19 = 2 \cdot 9 + 1$).

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**8** Let $x$ be the smallest positive integer satisfying the simultaneous congruences $x \equiv 1 \pmod 7$ (whose initial progression is marked on the number line) and $x \equiv 0 \pmod{11}$. Find $x$.

**a.** 29

**b.** 33

**c.** 55

**d.** 22 ✓

Write $x = 7k + 1$ and substitute into the second congruence: $7k + 1 \equiv 0 \pmod{11}$. Solving modulo 11 gives the unique residue modulo $7 \cdot 11 = 77$, yielding smallest positive solution $x = 22$ (check: $22 = 3 \cdot 7 + 1$ and $22 = 2 \cdot 11 + 0$).

**9** Given that $p = 11$ is prime, find the least non-negative remainder when $9!$ is divided by 11.

**a.** 10

**b.** 2

**c.** 1 ✓

**d.** $-3$

By Wilson's Theorem, $10! \equiv -1 \pmod{11}$. Factoring out the top 1 terms gives $10! = (10) \cdot 9! \equiv (-1)^1 1! \cdot 9! \equiv -1 \pmod{11}$. Inverting modulo 11 yields $9! \equiv 1 \pmod{11}$.

**10** Given that $p = 13$ is prime, find the least non-negative remainder when $10!$ is divided by 13.

**a.** 7

**b.** 6 ✓

**c.** 12

**d.** 2

By Wilson's Theorem, $12! \equiv -1 \pmod{13}$. Factoring out the top 2 terms gives $12! = (12)(11) \cdot 10! \equiv (-1)^2 2! \cdot 10! \equiv -1 \pmod{13}$. Inverting modulo 13 yields $10! \equiv 6 \pmod{13}$.

**11** Given that $p = 17$ is prime, find the least non-negative remainder when $13!$ is divided by 17.

**a.** 3 ✓

**b.** 14

**c.** 4

**d.** 16

By Wilson's Theorem, $16! \equiv -1 \pmod{17}$. Factoring out the top 3 terms gives $16! = (16)(15)(14) \cdot 13! \equiv (-1)^3 3! \cdot 13! \equiv -1 \pmod{17}$. Inverting modulo 17 yields $13! \equiv 3 \pmod{17}$.

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**12** Given that $p = 19$ is prime, find the least non-negative remainder when $17!$ is divided by $19$.

**a.** $18$

**b.** $2$

**c.** $-3$

**d.** $1$ ✓

By Wilson's Theorem, $18! \equiv -1 \pmod{19}$. Factoring out the top $1$ terms gives $18! = (18) \cdot 17! \equiv (-1)^1 1! \cdot 17! \equiv -1 \pmod{19}$. Inverting modulo $19$ yields $17! \equiv 1 \pmod{19}$.

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